INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD

# Linguistic Data 3
# Data Modeling in Indian Languages

## Assignment 3

Ishita
(Roll No: 2022114004)

Date: April 5, 2026

# Contents


**1 Introduction** **2**

**2 Dataset** **2**
2.1 Telugu Treebank . . . . . 2
2.2 Hindi Treebank . . . . . 2

**3 Dependency Distance** **3**

**4 Implementation** **3**

**5 Dependency Distance Statistics** **4**

**6 Dependency Relations** **4**
6.1 Dependency Relations Comparison Plot . . . . . 5

**7 Morphological Feature Analysis** **5**

**8 Morphological Feature Visualization** **6**
8.1 Feature Distribution . . . . . 6
8.2 Feature Comparison . . . . . 7
8.3 Feature Heatmap . . . . . 8
8.4 Feature Value Distributions . . . . . 8

**9 Statistical Testing** **10**

**10 Visualization of Results** **10**
10.1 Histogram Comparison . . . . . 10
10.2 Boxplot Comparison . . . . . 11
10.3 Density Plot . . . . . 11
10.4 Histogram . . . . . 12

**11 Experimental Data Analysis** **12**
11.1 Q1: RT Distribution . . . . . 12
11.2 Q2: Faceted Histograms . . . . . 13
11.3 Q3: Z-score Analysis . . . . . 13
11.4 Q4: Noun Frequency Distribution . . . . . 13
11.5 Q6: Noun vs Verb RT . . . . . 14
11.6 Q7: Mean RT with Confidence Intervals . . . . . 14
11.7 Q8: RT by Initial Letter . . . . . 15
11.8 Q9: Two Consonants Comparison . . . . . 15

**12 Discussion** **15**

**13 Conclusion** **16**

# 1 Introduction

This assignment explores structural properties of dependency trees in Indian languages. We perform a comparative analysis between Telugu and Hindi treebanks to understand differences in syntactic structure.

Dependency distance is an important concept in computational linguistics and psycholinguistics. It measures the linear distance between a head word and its dependent in a dependency tree.

Languages with freer word order may exhibit longer dependency distances, while languages with stricter word order tend to keep related words closer together.

In this assignment we:

- Loaded and processed Telugu and Hindi treebank data.
- Computed dependency distance statistics.
- Analyzed dependency relations.
- Examined morphological feature distributions.
- Performed statistical hypothesis testing.
- Visualized the distributions using plots.

The implementation was done using Python for data processing and R for statistical analysis and visualization.

# 2 Dataset

Two dependency treebanks were used.

## 2.1 Telugu Treebank

The Telugu data comes from the IIIT-Hyderabad Telugu treebank. The file used was:

```
iiit_hcu_intra_chunk_v1.conll
```

Total tokens:

```
17262
```

## 2.2 Hindi Treebank

Hindi data consists of multiple CoNLL formatted files. All files were read and merged into a single dataset.

Total tokens:

```
404058
```

# 3 Dependency Distance

Dependency distance is defined as the absolute difference between the position of a word and the position of its head.

$$distance = |index - head|$$

Example:

```
ID  FORM  HEAD
1   this  2
2   book  0
```

Distance between "this" and its head "book" is:

$$|1 - 2| = 1$$

This metric helps quantify syntactic complexity.

# 4 Implementation

The analysis was implemented in Python.

Project structure:

```
assignment3

   data
      telugu
         iiit_hcu_intra_chunk_v1.conll

      hindi
          (multiple .conll files)

   src
      parser.py
      dependency_analysis.py
      morphology_analysis.py
      stats_tests.py
      main.py
      test_load.py

   results
      dependency_boxplot.png
      dependency_density.png
      dependency_distance_histogram.png
      dependency_relations_comparison.png
      dependency_summary.csv
      hindi_morph_features.csv
      hindi_morph_features.png
```

```
    hindi_top_dependencies.csv
    morph_feature_comparison.csv
    morph_feature_comparison.png
    morph_feature_heatmap.png
    morph_value_case.png
    morph_value_gen.png
    morph_value_num.png
    morph_value_pers.png
    telugu_morph_features.csv
    telugu_morph_features.png
    telugu_top_dependencies.csv
    Q1_rt_histogram_all.png
    Q2_rt_histogram_faceted.png
    Q3_high_z_words.csv
    Q4_nounfreq_distribution.png
    Q6_boxplot_noun_verb.png
    Q7_barplot_ci_noun_verb.png
    Q8_boxplot_by_letter.png
    Q9_boxplot_two_consonants.png
  README.tex
```

The Python program loads the treebank, computes statistics, and generates plots.

To run the analysis:

```
python src/main.py
```

# 5 Dependency Distance Statistics

Results from the Python analysis:

```
Telugu Dependency Distance Stats
Mean: 1.56
Median: 1

Hindi Dependency Distance Stats
Mean: 3.19
Median: 1
```

Observation:

Hindi shows a larger mean dependency distance than Telugu. This suggests that syntactic dependencies in Hindi tend to span longer distances.

# 6 Dependency Relations

Top dependency relations in Telugu:

```
rsym
root
```

```
k1
nmod
k2
vmod
adv
k7p
k7t
r6
```

Top dependency relations in Hindi:

```
lwg__psp
nmod__adj
rsym
k1
ccof
pof__cn
r6
main
k2
lwg__vaux
```

Hindi uses several complex relation labels, reflecting morphological and syntactic information encoded in the treebank.

## 6.1 Dependency Relations Comparison Plot

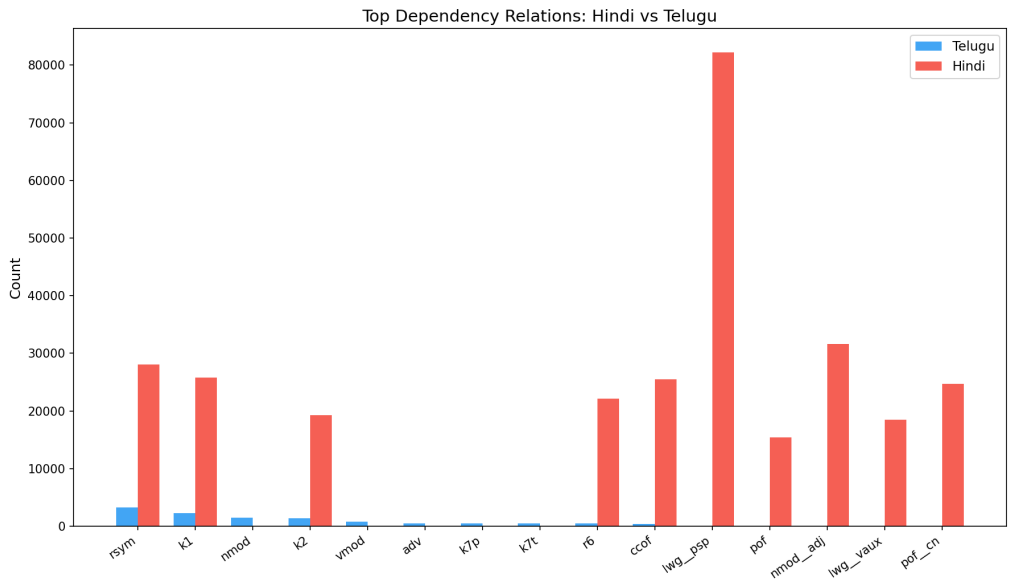


Figure 1: Comparison of top dependency relations in Hindi and Telugu

# 7 Morphological Feature Analysis

Both datasets contain rich morphological annotations.

Common features include:

- cat
- gen
- num
- pers
- case
- vib
- tam
- chunkId
- chunkType
- stype

These features describe grammatical categories such as gender, number, person, and tense-aspect-mood.

# 8 Morphological Feature Visualization

## 8.1 Feature Distribution

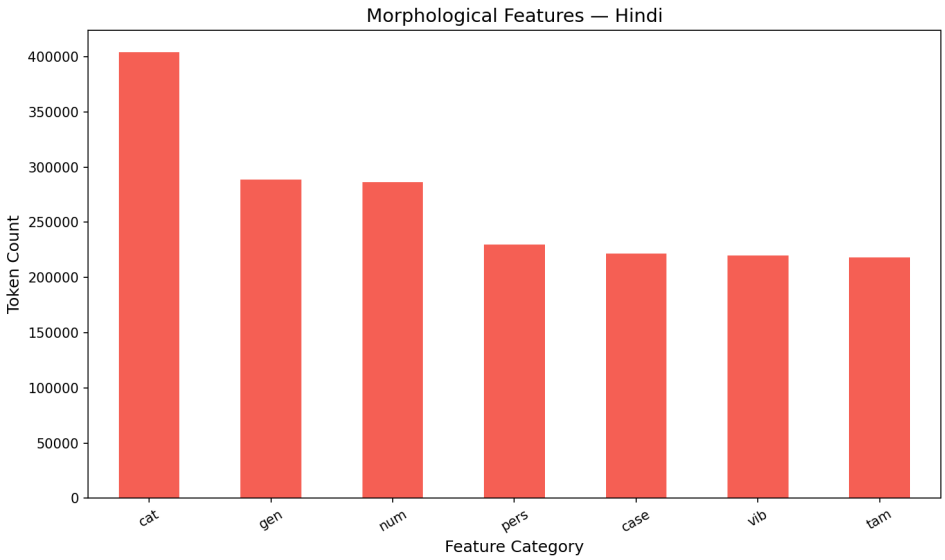


Figure 2: Morphological feature distribution in Hindi

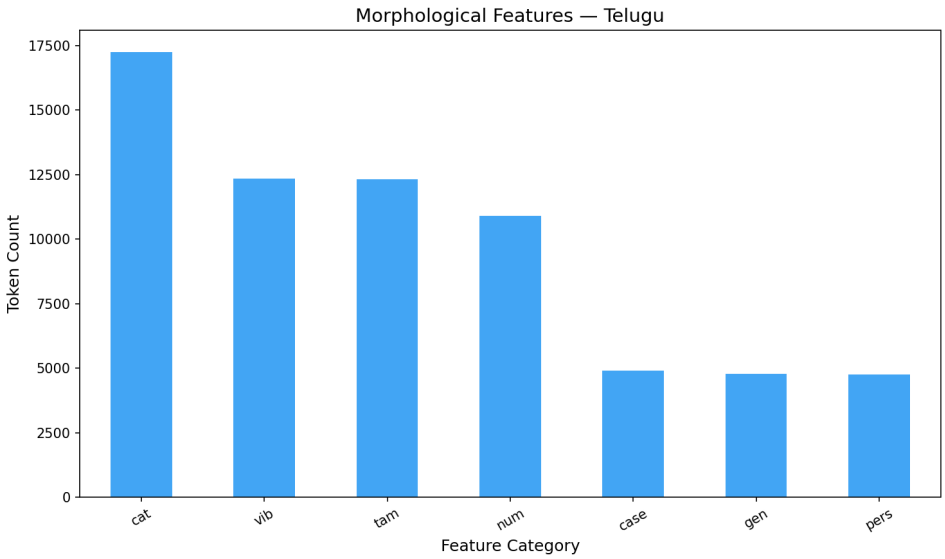


Figure 3: Morphological feature distribution in Telugu

## 8.2 Feature Comparison

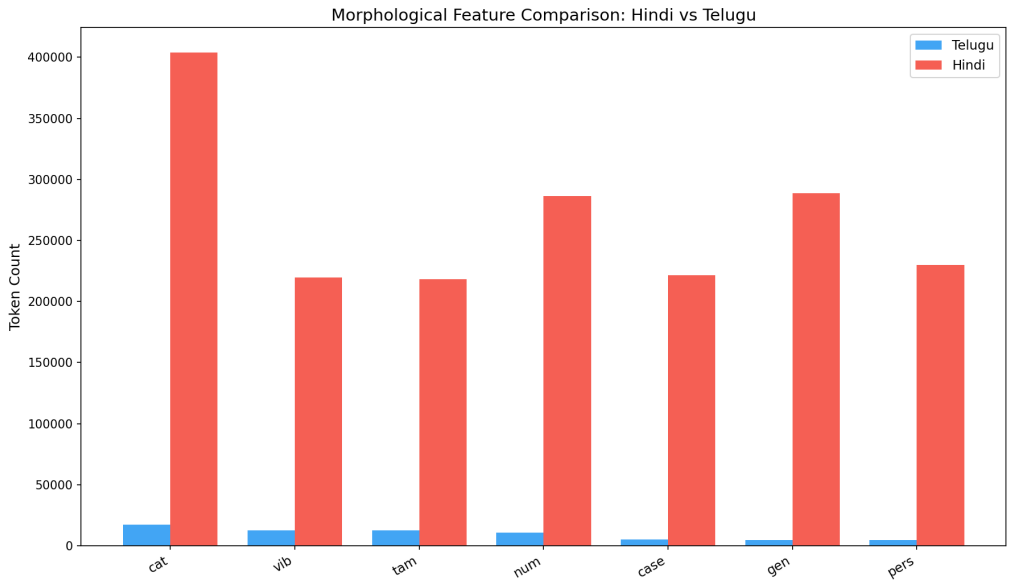


Figure 4: Comparison of morphological features across languages

## 8.3 Feature Heatmap

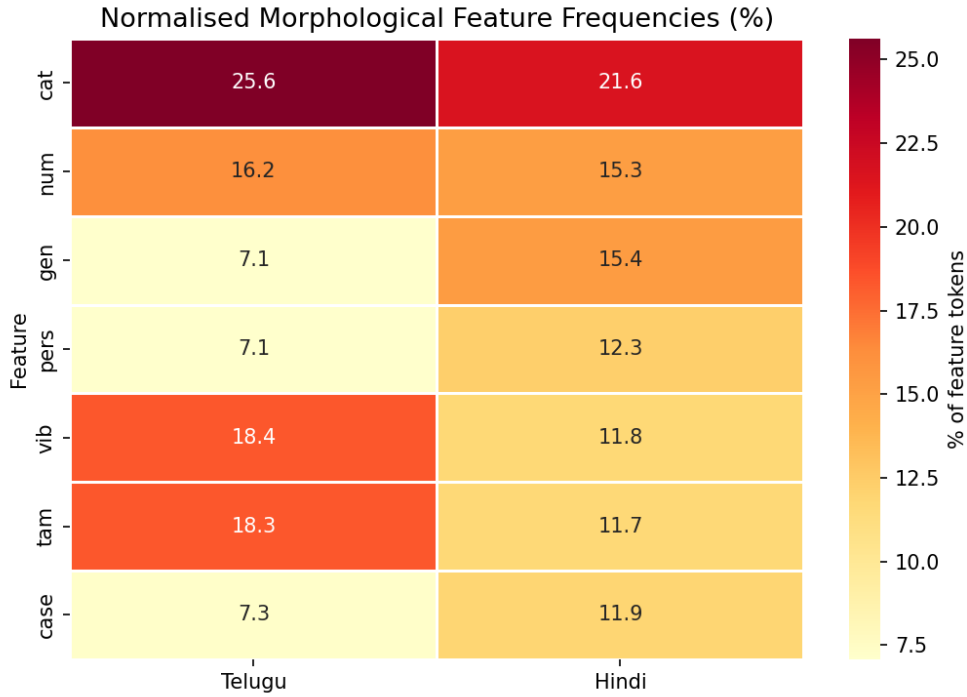


Figure 5: Heatmap of morphological feature frequencies

## 8.4 Feature Value Distributions

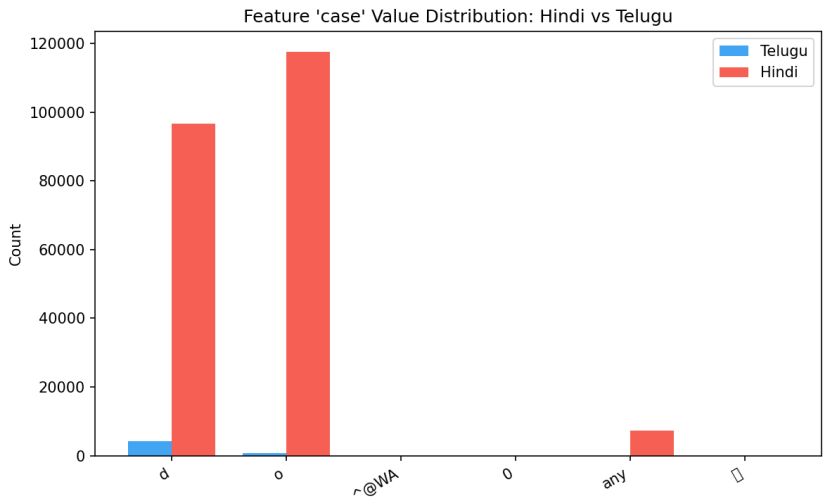


Figure 6: Distribution of Case values

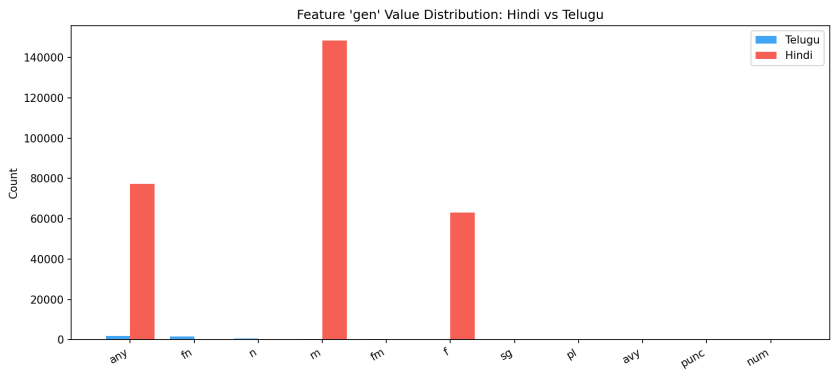


Figure 7: Distribution of Gender values

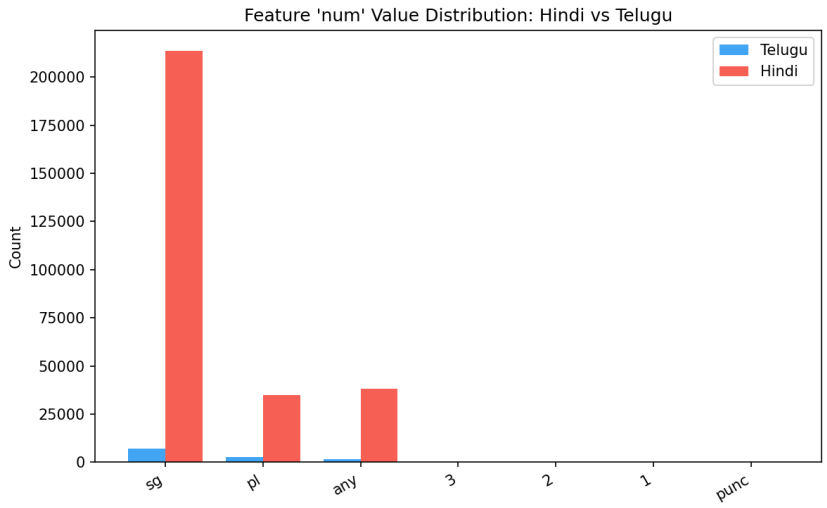


Figure 8: Distribution of Number values

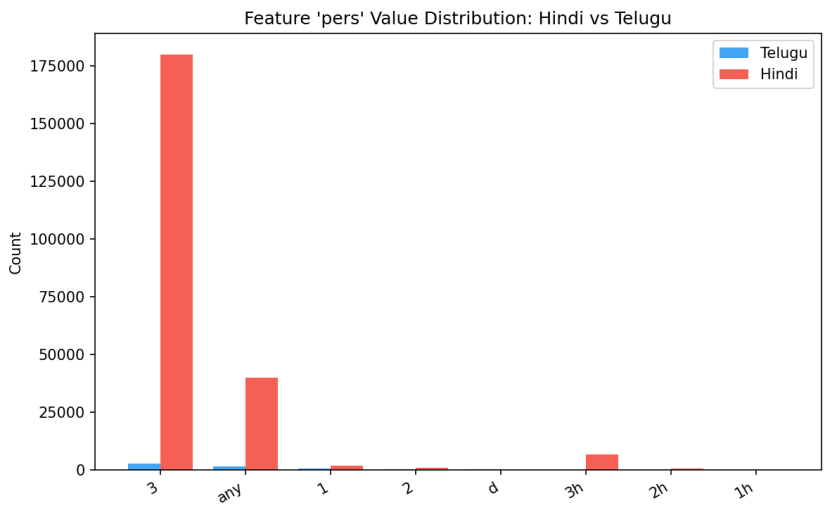


Figure 9: Distribution of Person values

# 9 Statistical Testing

A two-sample t-test was performed to determine whether dependency distances differ significantly between Telugu and Hindi.

```
t-statistic: -132.34
p-value: 0.0
```

Since:

$$p < 0.05$$

the difference is statistically significant.
This confirms that the two languages show different dependency distance patterns.

# 10 Visualization of Results

Several plots were generated to visualize the dependency distance distributions.

## 10.1 Histogram Comparison

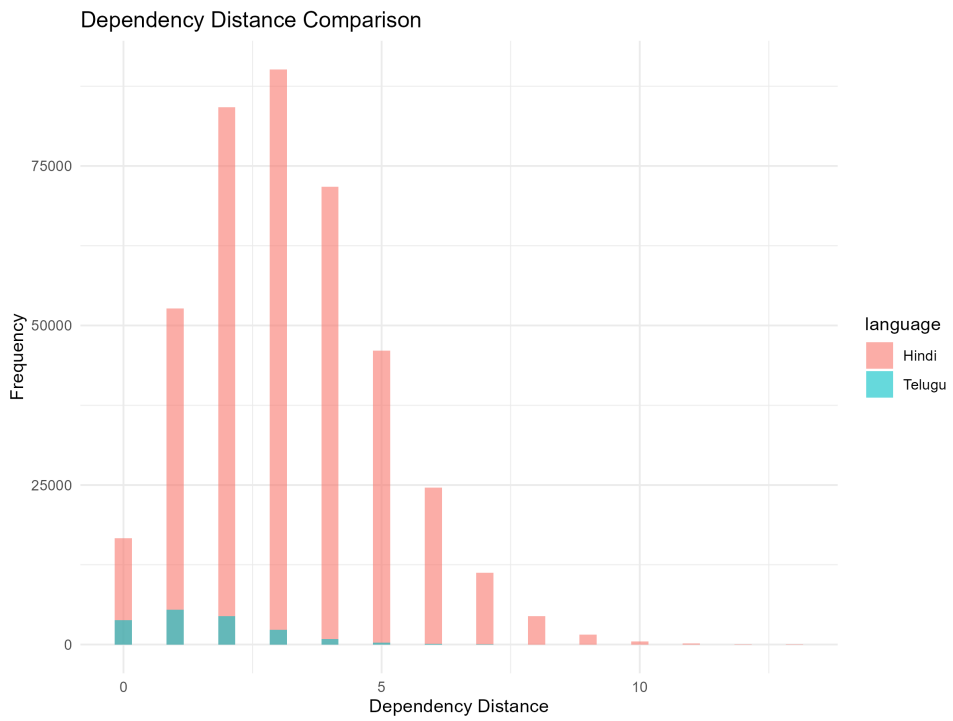


Figure 10: Histogram comparing dependency distances in Telugu and Hindi

## 10.2 Boxplot Comparison

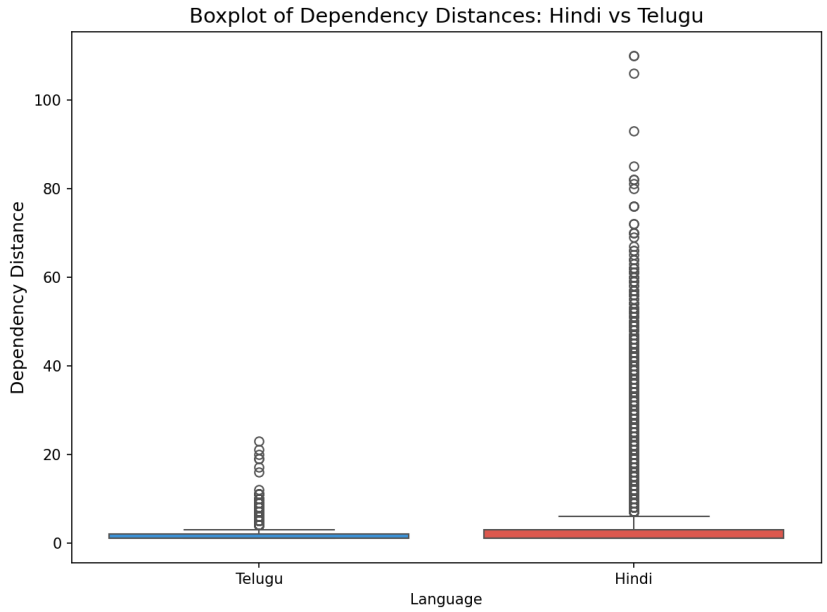


Figure 11: Boxplot showing dependency distance distributions

## 10.3 Density Plot

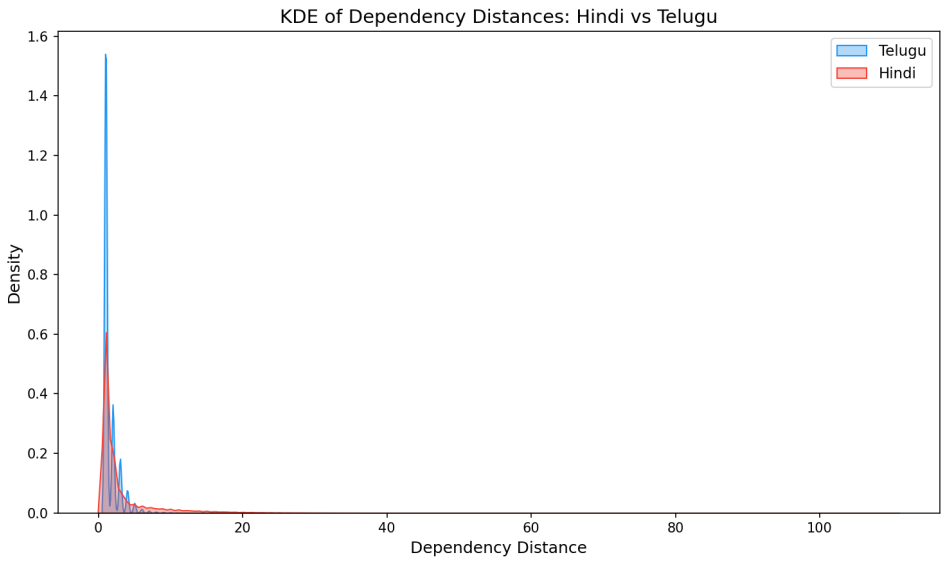


Figure 12: Density plot comparing the two languages

## 10.4 Histogram

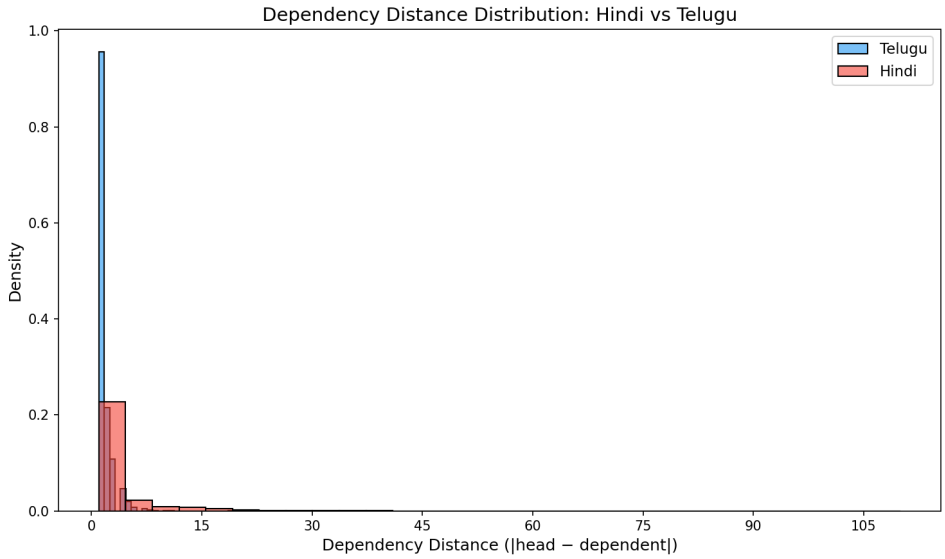


Figure 13: Dependency distance histogram generated from analysis

# 11 Experimental Data Analysis

## 11.1 Q1: RT Distribution

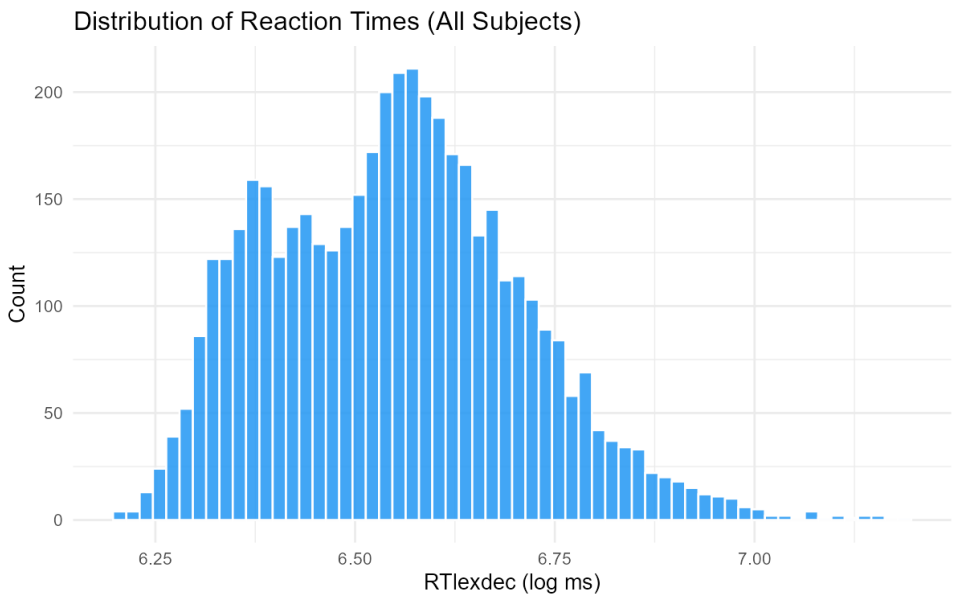


Figure 14: Histogram of reaction times (RT)

RT distribution is right-skewed and does not appear perfectly normal.

## 11.2 Q2: Faceted Histograms

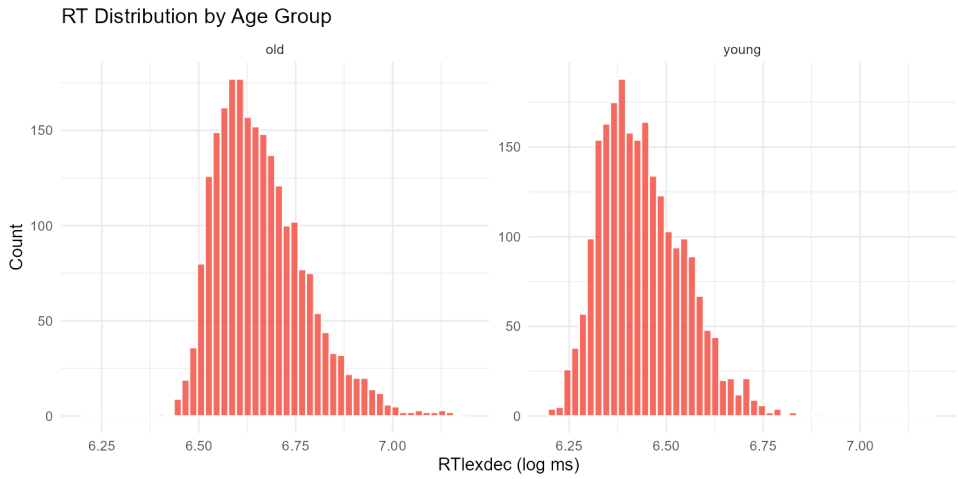


Figure 15: RT distribution for young vs old subjects

Young subjects show a more compact distribution compared to older subjects.

## 11.3 Q3: Z-score Analysis

High z-score words were extracted and stored in:

`Q3_high_z_words.csv`

These likely correspond to rare or complex words causing longer processing time.

## 11.4 Q4: Noun Frequency Distribution

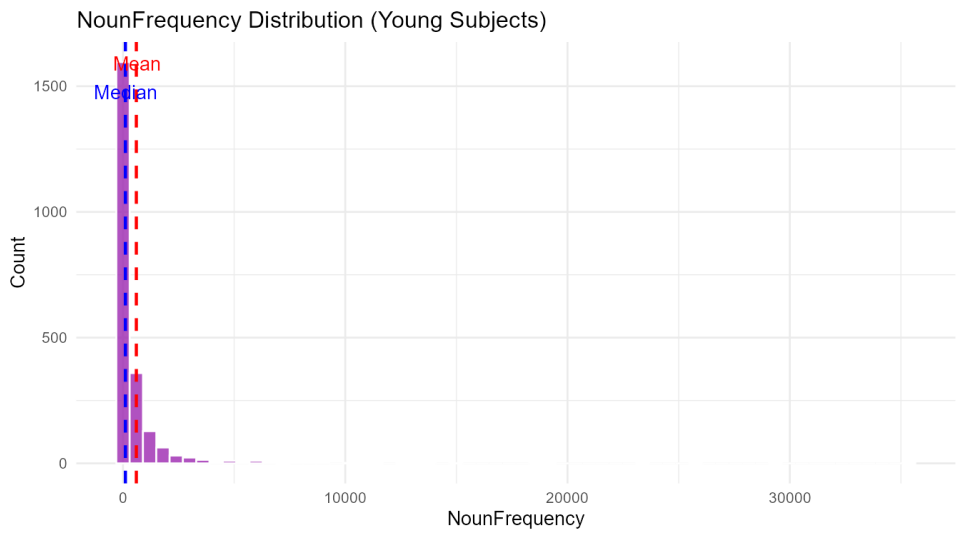


Figure 16: Distribution of noun frequency

The distribution is highly skewed, explaining differences between mean and median.

## 11.5 Q6: Noun vs Verb RT

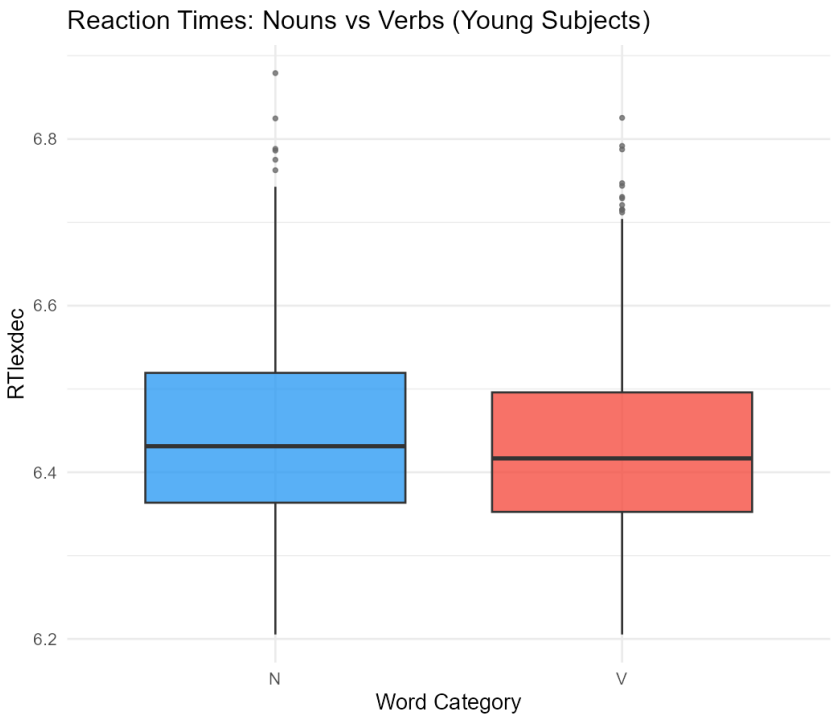


Figure 17: Boxplot comparing noun and verb RT

## 11.6 Q7: Mean RT with Confidence Intervals

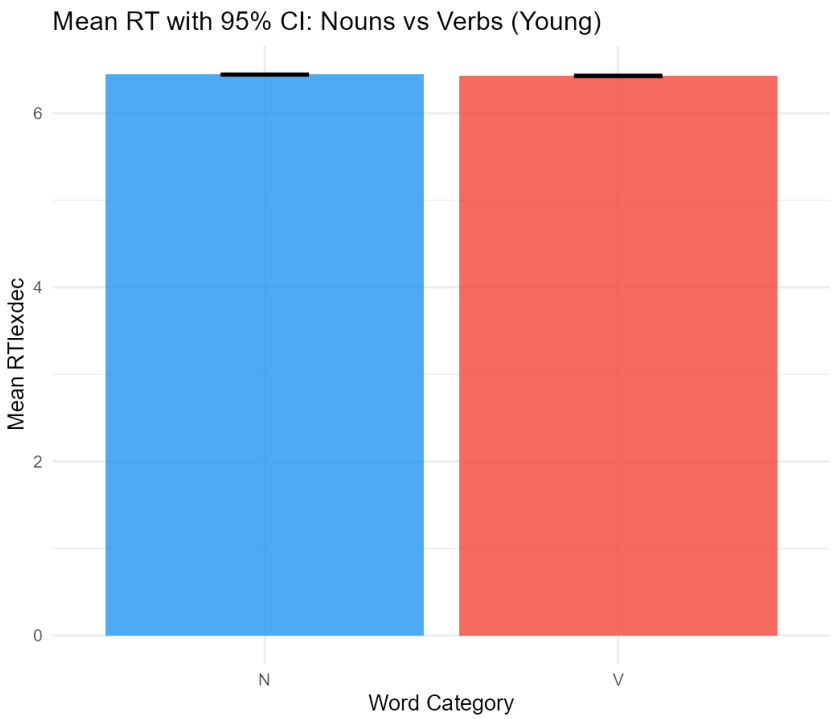


Figure 18: Mean RT with 95% confidence intervals

## 11.7 Q8: RT by Initial Letter

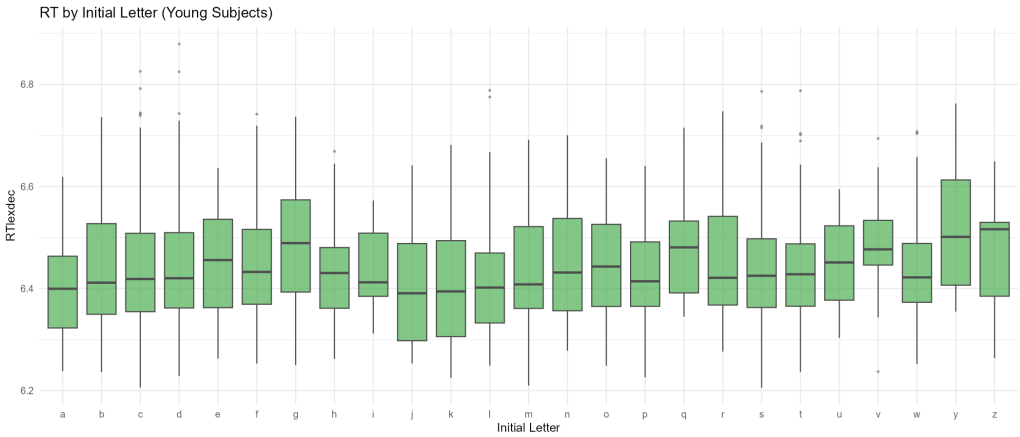


Figure 19: RT distribution by initial letter

## 11.8 Q9: Two Consonants Comparison

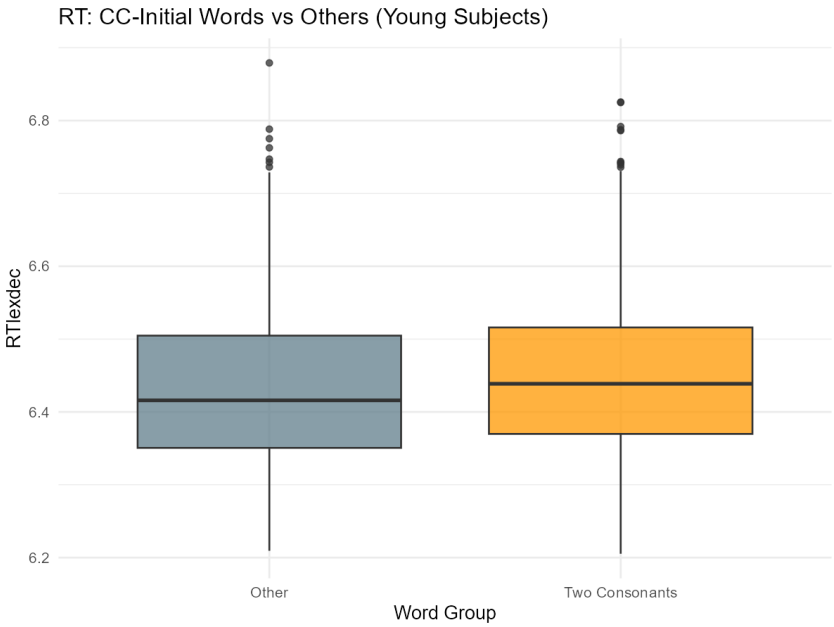


Figure 20: RT comparison for words starting with two consonants

Statistical testing suggests differences in RT depending on phonological structure.

# 12 Discussion

The analysis reveals several interesting observations:

- Hindi exhibits larger average dependency distances compared to Telugu.
- Telugu dependencies tend to be shorter and more local.

- Morphological annotation structures are similar across the datasets.
- Dependency relations reflect the syntactic structure of each language.

These findings align with linguistic observations that Indian languages often exhibit flexible word order but still maintain strong syntactic dependencies.

# 13 Conclusion

In this assignment we analyzed dependency treebanks for Telugu and Hindi.

We implemented a pipeline that loads treebanks, extracts syntactic statistics, and performs comparative analysis.

Key findings include:

- Hindi has larger average dependency distances.
- The difference between the languages is statistically significant.
- Morphological annotation is extensive in both treebanks.
- Experimental data analysis showed that RT distributions are skewed and influenced by lexical and phonological factors.

This type of analysis helps us understand structural properties of languages and provides insights into syntactic complexity.

Such metrics are useful in computational linguistics, psycholinguistics, and natural language processing.

**Code Repository:**

`https://github.com/ishitab1310/Linguistic-Data-3-Data-Modeling-in-ILs/tree/main/assignment3`